Downstream Benchmarks of Pareto-Optimal Compressed Models: Qwen2.5-0.5B-Instruct

Rasa Khosrowshahli

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Abstract

Two multi-objective evolutionary searches (NSGA-II) compress Qwen2.5-0.5B-Instruct. The first (UBQ-ECSQ) trades calibration loss against encoded storage size, with global quantization and pruning parameters; the second (hardware-aware) trades calibration loss against predicted GPU decode latency, with a per-layer-group choice among bf16 and int4 kernel formats, MLP pruning and sublayer removal. From each Pareto front, the solutions at calibration-loss increases of 0%, 1%, 2%, 5% and 10% are evaluated, with the uncompressed model, on eight standard benchmarks: six zero-shot commonsense and reasoning tasks (PIQA, ARC-Easy, ARC-Challenge, HellaSwag, WinoGrande, BoolQ), 5-shot MMLU and 5-shot GSM8K. UBQ-ECSQ: at 8.0 bits per weight the model is 2.00× smaller with no loss of accuracy; at 6.0–6.7 bits per weight (2.39–2.66× smaller) it keeps 98% of the average accuracy; at 5.0 bits per weight (3.2× smaller) it keeps 88%. Hardware-aware: 1.56× faster decoding keeps 98.8% of the accuracy, and 1.98× faster decoding with 3.4× less memory (4.5 bits per weight, almost all int4) keeps 95.1%; removing sublayers costs far more accuracy than it saves time.

1 UBQ-ECSQ compression scheme

All 169 weight matrices of the model are compressed, including the token embedding, which is tied to the output head (494.0M weights, 100% of the parameters). Every matrix uses the same five parameters (genes). For a weight w in output row r, with σ_r and $m_r = \max|w_r|$ taken from the original weights:

1. **Pruning:** $w \to 0$ if $\alpha\,\sigma_r \le w \le \beta\,\sigma_r$, with $\alpha \le 0 \le \beta$.
2. **Uniform quantization (UBQ):** a symmetric, zero-preserving grid of k levels with step $s_r = c\,m_r/\lfloor (k-1)/2 \rfloor$, clipping factor $c \in [0.7, 1]$; pruned weights stay 0.
3. **Entropy-constrained merge (ECSQ):** each non-empty level gets a model-wide centroid; contiguous levels are merged by dynamic programming to minimize $D + \lambda R$ (added squared error plus entropy of level occupancy), keeping a fraction ρ of the levels. The zero level is never merged. $\rho = 1$ means no merge.

Encoded size is the smaller of a dense code ($\lceil \log_2 k_{\text{used}} \rceil$ bits per weight) and a bitmap code (1 bit per weight plus a code per non-zero), plus a 16-bit scale per row and a lookup table per matrix. Sizes are in MB (10^6 bytes); bits per weight include scales and tables.

2 UBQ-ECSQ search

- **Objectives (minimized):** $\Delta\mathcal{L}$, the next-token cross-entropy on the calibration set minus that of the uncompressed model, and the encoded size.

- **Constraint:** $\Delta\mathcal{L} \leq 1.0$ nats.
- **Calibration set:** 8 sequences of 2048 tokens from C4 (train, English), fixed for the whole run. Uncompressed loss $\mathcal{L}_0 = 2.8854$ nats.
- **Genes:** $\log_2 k$ with $k \in [16, 256]$ (range set by a warm-up on the model's error-vs-k curve), $c \in [0.7, 1]$, $\alpha \in [-1, 0]$, $\beta \in [0, 1]$, $\rho \in [0, 1]$.
- **NSGA-II:** population 100, 50 generations, seed 42.

3 UBQ-ECSQ Pareto fronts

Figure 1 shows the final front on the objective of the search, the calibration loss. Figure 2 scores the same 100 solutions on held-out text: perplexity on the full WikiText-2 test split and on 256 windows of C4 validation, with 2048-token windows. The test fronts follow the calibration front: the loss stays near the uncompressed model's down to about 370 MB and rises steeply below 310 MB.

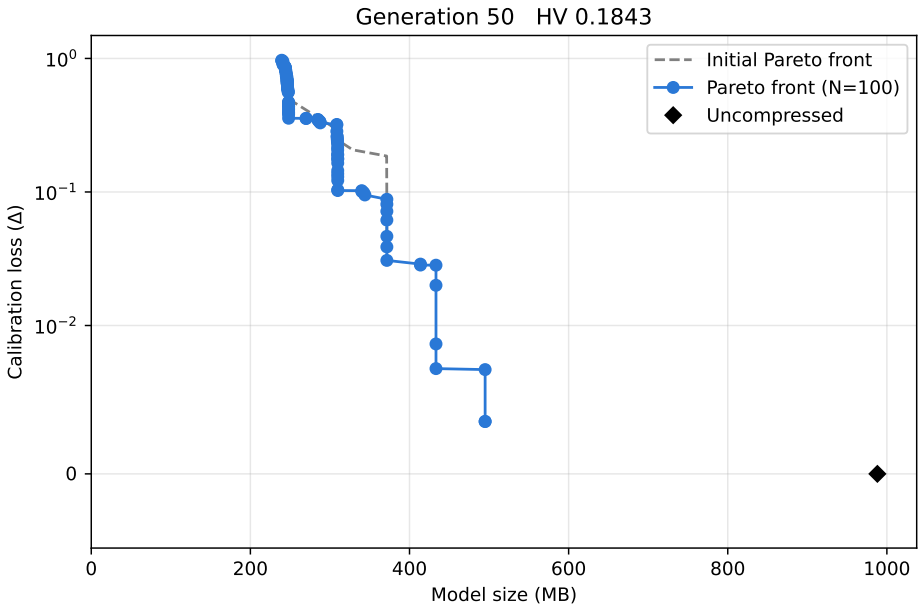
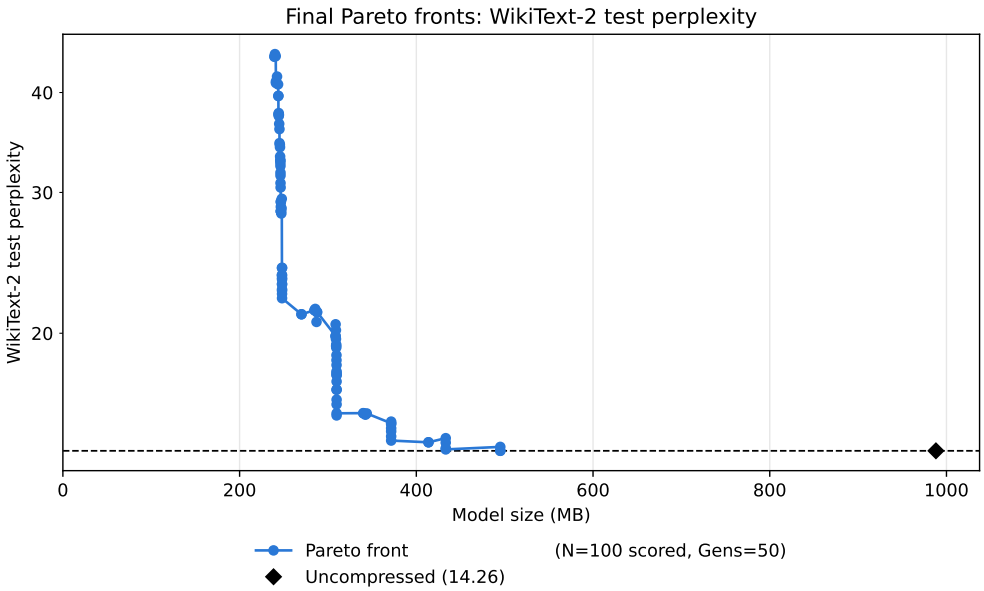
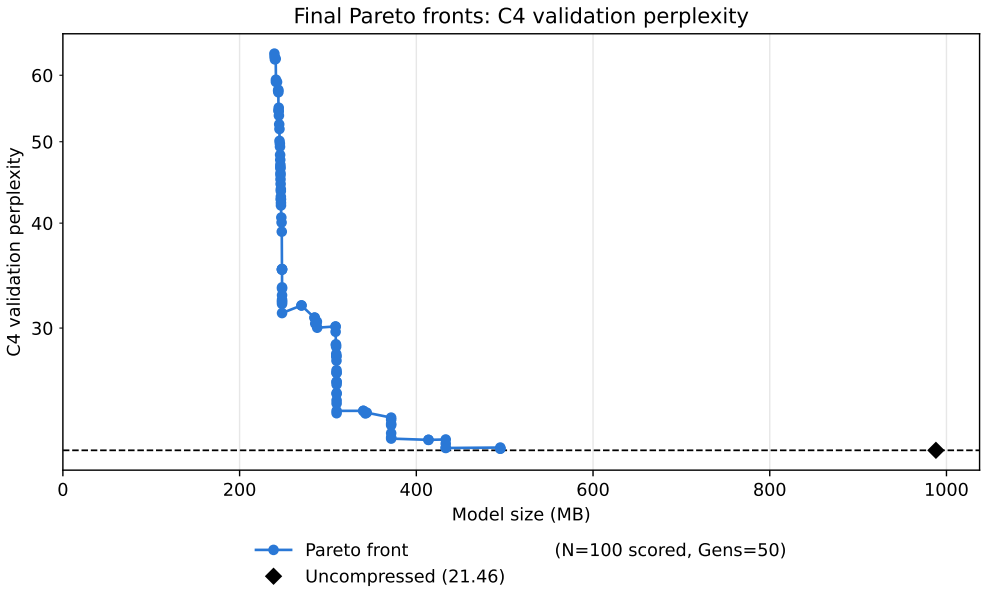


Figure 1: Final Pareto front (100 solutions) of calibration $\Delta\mathcal{L}$ against encoded size, with the front of the initial population and the uncompressed model.



(a) WikiText-2 test



(b) C4 validation

Figure 2: Test perplexity of the final Pareto front against encoded size. The dashed line is the uncompressed model.

4 Selected UBQ-ECSQ models

The calibration error of a solution is $\Delta\mathcal{L}/\mathcal{L}_0$. For each level (0, 1, 2, 5, 10%) the smallest solution within it is selected. No solution has zero error, so the most accurate one (0.12%) stands for the 0% level. Every solution smaller than 309 MB exceeds 10% error, so the 5% and 10% models are nearly the same size but reach it with different settings: Q10 prunes more ($\beta = 0.32$ against 0.02) and clips harder ($c = 0.85$ against 0.93).

Model	k	c	Sparsity (%)	Calib. err. (%)	Size (MB)	Smaller by	Bits/w	Huffman bits/w	WikiText-2 ppl	C4 ppl
Uncompressed (bf16)	–	–	0.0	0.00	988	1.00×	16.00	–	14.26	21.46
Q0 (0%)	164	0.995	2.2	0.12	495	2.00×	8.02	6.47	14.26	21.55
Q1 (1%)	65	0.934	5.3	1.00	414	2.39×	6.70	5.23	14.61	22.08
Q2 (2%)	52	0.949	6.8	1.61	372	2.66×	6.01	4.84	15.08	22.49
Q5 (5%)	29	0.928	11.8	4.75	310	3.19×	5.01	4.05	17.00	25.05
Q10 (10%)	28	0.849	19.2	9.87	309	3.20×	5.00	3.98	20.52	29.70

Table 1: Selected solutions. Sparsity counts pruned weights and weights rounded to zero. Huffman bits/w is the storage with Huffman-coded codes (reported only, not optimized). None of the five merges levels ($\rho = 1$).

5 Benchmark protocol

All models are evaluated with the same harness (lm-evaluation-harness 0.4.13), in bf16, on full evaluation splits, without a chat template, with fixed seeds. Compressed weights are reconstructed in place, so the models differ only in their weight values. The hardware-aware models were evaluated on a second machine together with their own uncompressed reference, which agrees with the first within one standard error on every task; each model is compared with the reference from its own machine.

Benchmark	What it tests	Split	Questions	Shots	Metric
PIQA	Physical commonsense: the right way to do an everyday task	validation	1,838	0	acc. (norm.)
ARC-Easy	Grade-school science questions	test	2,376	0	acc. (norm.)
ARC-Challenge	Harder grade-school science questions (retrieval-resistant)	test	1,172	0	acc. (norm.)
HellaSwag	Commonsense: the most plausible continuation of an everyday scene	validation	10,042	0	acc. (norm.)
WinoGrande	Commonsense pronoun resolution (fill in the blank, two options)	validation	1,267	0	accuracy
BoolQ	Reading comprehension: yes/no questions about a Wikipedia passage	validation	3,270	0	accuracy
MMLU	Knowledge and reasoning in 57 subjects (STEM, humanities, law, medicine, ...)	test	14,042	5	accuracy
GSM8K	Multi-step grade-school math word problems, solved in writing	test	1,319	5	exact match

Table 2: Benchmarks. Multiple-choice tasks pick the answer with the highest log-likelihood (acc. (norm.): log-likelihood normalized by answer length); GSM8K generates a solution and compares the extracted final number with the reference.

6 UBQ-ECSQ results

Model	Size (MB)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU 5-shot	GSM8K 5-shot	Average
Uncompressed	988	70.4	59.2	33.1	52.4	56.0	67.8	47.0	31.2	52.1
Q0 (0%)	495	70.2	59.9	33.5	52.4	56.6	68.4	47.0	31.7	52.5
Q1 (1%)	414	70.2	59.6	33.7	52.5	55.9	64.3	46.5	27.6	51.3
Q2 (2%)	372	69.5	61.0	32.8	52.0	54.2	66.6	46.1	26.7	51.1
Q5 (5%)	310	69.2	55.2	32.0	50.0	56.4	49.7	45.0	9.7	45.9
Q10 (10%)	309	67.5	56.2	31.0	47.7	57.2	56.1	38.3	12.6	45.8
Std. error		1.1	1.0	1.4	0.5	1.4	0.8	0.4	1.3	

Table 3: Accuracy (%, higher is better), with the unweighted average over the eight benchmarks. Last row: standard error of the uncompressed model's accuracy.

Model	Size (MB)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU	GSM8K	Avg. kept (%)
Q0 (0%)	495	−0.2	+0.7	+0.4	−0.1	+0.6	+0.6	0.0	+0.5	100.6
Q1 (1%)	414	−0.2	+0.5	+0.6	0.0	−0.2	−3.5	−0.4	−3.6	98.4
Q2 (2%)	372	−0.9	+1.9	−0.3	−0.5	−1.8	−1.2	−0.8	−4.5	98.0
Q5 (5%)	310	−1.2	−4.0	−1.1	−2.4	+0.3	−18.1	−2.0	−21.5	88.0
Q10 (10%)	309	−2.9	−2.9	−2.1	−4.7	+1.2	−11.8	−8.6	−18.6	87.9

Table 4: Change in accuracy against the uncompressed model (percentage points; negative is a loss). Avg. kept is the compressed model's average accuracy as a percentage of the uncompressed model's.

Observations.

- **8 bits per weight is lossless.** Q0 is 2.00× smaller and every benchmark is within 0.7 points of the uncompressed model, inside the standard errors (100.6% kept).
- **1–2% calibration error keeps 98% of the accuracy.** Q1 and Q2 are 2.39× and 2.66× smaller. The commonsense tasks and MMLU stay within 2 points; GSM8K drops 3.6–4.5 points and BoolQ up to 3.5.
- **At 5.0 bits per weight, about 12% of the average accuracy is lost.** Q5 and Q10 are 3.2× smaller and keep 88%. The commonsense tasks lose at most 5 points; BoolQ breaks (Q5: 49.7%, below the 62.2% of always answering "yes").
- **GSM8K is the most sensitive benchmark:** 31.2% uncompressed, 27–28% at 1–2% error, 10–13% at 5–10% error.
- **Calibration loss does not rank the models downstream.** Q5 has half the calibration error of Q10 and lower test perplexity, and is better on MMLU (45.0 against 38.3), but worse on BoolQ and GSM8K; the two averages are equal. Below about 6 bits per weight, models need a benchmark check before use.

7 Hardware-aware compression

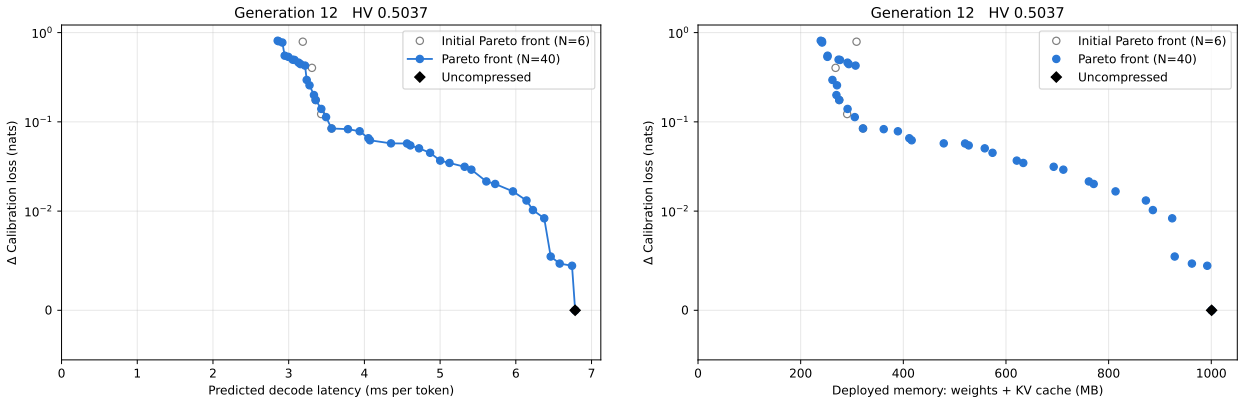
7.1 Scheme and search

The compressed model is the deployed model: every choice maps onto a kernel that runs on the GPU, and the cost is the decode latency and memory of that model, not a storage size.

- **Formats:** bf16, or int4 with an asymmetric bf16 scale and zero per group of 32, 64 or 128 input weights (g32, g64, g128), on PyTorch's int4 matrix-multiplication kernel. The clipping of each int4 group minimizes the activation-weighted error and is solved per group, not searched.
- **Genes (23):** the format of each of four roles (attention q/k/v, attention output, MLP up/gate, MLP down) in each of four depth bands of 6 layers (16 genes); the format of the output head, tied to the token embedding (1); the fraction of MLP neurons kept per band, from $\{1, 0.875, 0.75, 0.625, 0.5\}$, by activation-weighted importance (4); the number of attention and of MLP sublayers removed, 0–6 each, in a fixed order ranked by the loss increase of removing each sublayer alone (2).
- **Objectives (minimized):** $\Delta\mathcal{L}$ on the same calibration set as above ($\mathcal{L}_0 = 2.8854$ nats), and the decode latency of one token (batch 1, 1024 tokens of context), predicted from per-kernel times measured on an RTX 3060 GPU. Memory (weights plus the KV cache of 1024 tokens; 1001 MB uncompressed) is reported, not optimized.
- **Constraint:** $\Delta\mathcal{L} \leq 1.0$ nats. **NSGA-II:** population 40, 12 generations, seed 42.

7.2 Pareto fronts

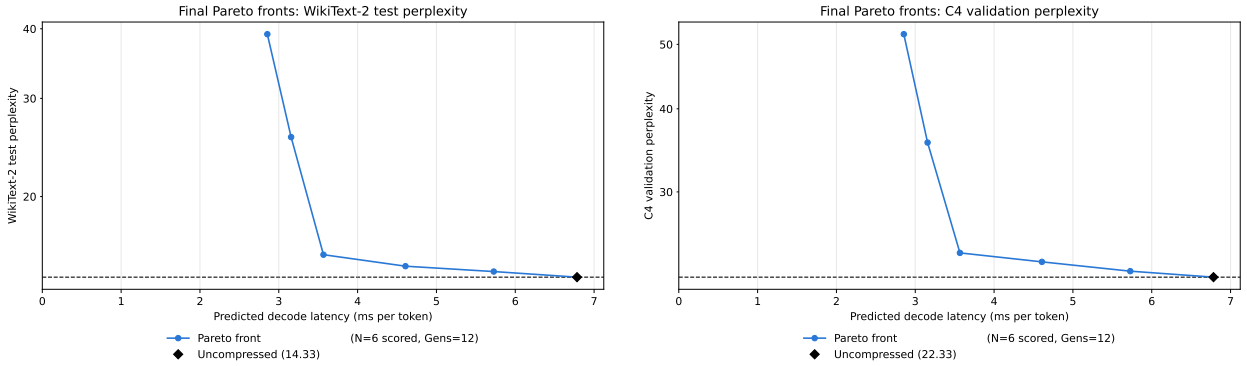
The final front has 40 solutions from 2.85 to 6.78 ms and 239 to 1001 MB (Figure 3). With formats alone, $\Delta\mathcal{L}$ stays at or below 0.18 nats (6%) down to 3.36 ms, the model with every matrix in int4 g128; every faster solution prunes MLP neurons or removes sublayers, and $\Delta\mathcal{L}$ rises to 0.20–0.81 nats. Five solutions spread along the front and the uncompressed model were scored on 32 windows each of WikiText-2 test and C4 validation (Figure 4).



(a) Against predicted decode latency (objective)

(b) Against deployed memory (reported)

Figure 3: Final Pareto front (40 solutions) of calibration $\Delta\mathcal{L}$, with the front of the initial population and the uncompressed model.



(a) WikiText-2 test

(b) C4 validation

Figure 4: Test perplexity of five solutions of the final front and the uncompressed model against predicted decode latency.

7.3 Selected models

For each level (0, 1, 2, 5, 10%) the fastest solution within it is selected (H0–H10). As above, no solution has zero error, so the most accurate one (0.16%) stands for the 0% level.

Model	Calib. err. (%)	Decode (ms)	Faster by	Memory (MB)	Smaller by	Bits/w	int4 (%)	Formats q/k/v out	up	down	head	Removed attn./MLP
Uncompressed	0.00	6.78	1.00×	1001	1.00×	16.00	0	BBBB	BBBB	BBBB	B	0/0
H0 (0%)	0.16	6.74	1.01×	992	1.01×	15.86	1	B3BB	BBBB	BBBB	B	0/0
H1 (1%)	0.75	5.61	1.21×	761	1.31×	12.13	35	BBBB	633B	B36B	B	0/0
H2 (2%)	1.99	4.35	1.56×	479	2.09×	7.55	77	33B3	B333	33B3	3	0/0
H5 (5%)	4.84	3.43	1.98×	291	3.43×	4.51	99	3B38	3386	8888	8	0/0
H10 (10%)	8.90	3.28	2.07×	271	3.70×	4.19	97	8836	6888	8888	8	1/1

Table 5: Selected hardware-aware solutions. Formats: one character per depth band (bands 1–4) for attention q/k/v, attention output, MLP up/gate and MLP down, then the output head; B = bf16, 3/6/8 = int4 with groups of 32/64/128. Bits/w counts weights only; memory includes the 13 MB KV cache. None prunes MLP neurons.

7.4 Results

Model	Decode (ms)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU 5-shot	GSM8K 5-shot	Average
Uncompressed	6.78	70.6	59.2	33.0	52.4	55.2	67.7	47.0	30.3	51.9
H0 (0%)	6.74	70.7	59.6	33.4	52.0	55.6	66.9	46.5	31.1	52.0
H1 (1%)	5.61	70.1	60.0	32.7	51.4	56.0	67.6	46.7	27.9	51.5
H2 (2%)	4.35	69.7	59.7	32.8	51.0	56.7	66.1	45.9	28.5	51.3
H5 (5%)	3.43	69.0	58.0	31.5	49.4	55.0	66.5	44.8	20.5	49.4
H10 (10%)	3.28	67.3	49.7	29.0	46.4	52.4	63.8	37.6	4.9	43.9
Std. error		1.1	1.0	1.4	0.5	1.4	0.8	0.4	1.3	

Table 6: Accuracy (%, higher is better) of the hardware-aware models, with the unweighted average.

Model	Decode (ms)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU	GSM8K	Avg. kept (%)
H0 (0%)	6.74	+0.2	+0.4	+0.3	−0.4	+0.4	−0.9	−0.5	+0.8	100.1
H1 (1%)	5.61	−0.5	+0.8	−0.3	−0.9	+0.9	−0.1	−0.3	−2.4	99.3
H2 (2%)	4.35	−0.8	+0.5	−0.3	−1.4	+1.5	−1.6	−1.1	−1.8	98.8
H5 (5%)	3.43	−1.5	−1.1	−1.5	−2.9	−0.2	−1.2	−2.2	−9.8	95.1
H10 (10%)	3.28	−3.3	−9.5	−4.0	−5.9	−2.8	−3.9	−9.4	−25.4	84.6

Table 7: Change in accuracy against the uncompressed model (percentage points; negative is a loss).

Observations.

- **The 0% level brings no speed-up.** H0 changes one band of one role to int4 (1% of the weights) and is 1.01× faster; any larger change costs measurable calibration loss.
- **1–2% calibration error: up to 1.56× faster at 98.8% accuracy.** H1 and H2 stay within 1.6 points on every benchmark except GSM8K (−2.4 and −1.8).
- **Almost-all-int4 is the efficient point.** H5 is 1.98× faster with 3.43× less memory and keeps 95.1%: the commonsense tasks and MMLU lose at most 2.9 points, GSM8K 9.8.
- **Removing sublayers does not pay.** H10 removes one attention and one MLP sublayer for 0.15 ms (4%) over H5 and loses another 5.5 points of average accuracy (84.6% kept); GSM8K falls to 4.9%, ARC-Easy and MMLU by 9.5 and 9.4 points.
- **Calibration loss ranks these models correctly:** accuracy falls monotonically with the level, unlike the UBQ-ECSQ models at 5–10%.
- **Against UBQ-ECSQ:** H5 at 4.5 bits per weight keeps 95.1%, the UBQ-ECSQ models at 5.0 bits per weight 88% (BoolQ 66.5 against 49.7, GSM8K 20.5 against 9.7). The UBQ-ECSQ sizes are entropy-coded storage that must be decoded before use; the hardware-aware models run directly on int4 kernels.